

Jorge Luis Castro García

jorgeca003@gmail.com | +506 8863 7483 | Heredia, Costa Rica | [linkedin.com/in/jlcastrog/](https://www.linkedin.com/in/jlcastrog/) | montezumalabs.com/

Professional Summary

Senior Software Engineer with 9+ years of experience delivering high-impact, scalable solutions for the semiconductor and high-tech industries. Proven track record in designing and maintaining complex systems that integrate hardware and software, accomplished through the creation of detailed feature specifications, workflow automation, and strategic coordination with R&D, manufacturing, and business stakeholders to gather requirements and deliver results. Recognized for high-level technical leadership and owning end-to-end feature development in ambiguous environments. Adept at bridging technical complexity with business while continuously identifying opportunities to optimize processes and modernize legacy systems.

Core Competencies

Software quality assurance, software verification and validation (V&V), Linux Real-Time targets, new product introduction (NPI) and factory test enablement, test automation and management, CI/CD, regulatory compliance (SBOM/vulnerabilities, threat modeling, cybersecurity requirements), requirements gathering and feature specification, hardware and software troubleshooting, agile and waterfall SDLC, standardization and documentation of processes, architecture and design review, cross-functional stakeholder management, continuous improvement, legacy system modernization, mentoring and coaching.

Experience

October 2023 – Present: Full Stack Software Engineer (National Instruments – Emerson T&M)

- Lead initiatives to align STS firmware update processes with EU Cyber Resilience Act (CRA) requirements, ensuring regulatory compliance and strengthening system security for European market readiness.
- Served as technical lead for two business-critical internal tools, driving architecture decisions, code reviews, and feature delivery while coordinating with cross-functional teams to ensure scalability, reliability, and alignment with product requirements.
- Designed and implemented an AI-powered diagnostic solution using .NET 9.0, LLMs, and Retrieval-Augmented Generation (RAG) to analyze system logs, drastically reducing manual troubleshooting time.
- Designed and maintained backend services and frontend applications using C#, MySQL, JavaScript, Python, and LabVIEW, supporting monitoring and maintenance platforms for semiconductor test systems used in high-reliability manufacturing environments.
- Contributed key features to the Semiconductor Test System (STS) Maintenance Software, a C#-based platform for monitoring and maintaining system health, including integration of in-development hardware devices to expand diagnostic and calibration capabilities.
- Achieved an 80% reduction in manual hardware data entry by developing a Python-based automation tool for R&D libraries, significantly improving data reliability.
- Collaborated with cross-functional global teams (Sales, Manufacturing, and R&D) to drive continuous improvement in an iterative development environment, ensuring alignment across product lifecycle stages.

November 2019 – October 2023: Technical Support Engineer (National Instruments – Emerson T&M)

- Diagnosed and resolved complex hardware-software integration issues, performing root-cause analysis and collaborating with R&D, Sales, and Product Planning to deliver short and long-term fixes.
- Provided technical support for NI data acquisition systems (PXI, cDAQ, and Linux Real-Time targets) using diverse programming languages, including C#, Python, C/C++, and LabVIEW.
- Acted as Subject Matter Expert for key software platforms: LabVIEW, LabWindows/CVI (ANSI C), Measurement Studio (C#/VB.NET), driving knowledge-sharing and mentoring.

- Developed LabVIEW Connectivity Learning Path, a comprehensive course on building robust network communication and data integration solutions with LabVIEW.
- Trained support engineers in C# and ANSI C, expanding team proficiency in text-based programming and improving customer support for complex measurement and automation projects.
- Maintained a 4.8/5.0 Customer Satisfaction (CSAT) average, earning recognition for high-quality resolution of complex system-level escalations. Exceeded productivity benchmarks with an efficiency of 3.14 hr/case, 28% faster than the department average.

January 2017 – November 2019: Test Development Engineer (Teradyne)

- Designed and executed diagnostic, regression, and performance tests for semiconductor platforms (C#, C++, VB.NET).
- Improved engineering efficiency by building a Visual Studio extension that automated test project creation through customized templates.
- Automated semiconductor test workflows by designing and building a C#-based results management application with integrated server uploads and real-time user notifications.
- Resolved internal and external customer issues with core software architecture and user interface, ensuring timely solutions by collaborating with local and US-based teams using Agile methodologies.

March – June 2016: Intern (Continental AG, Germany)

- Conducted geometric verification of valve blocks in vehicle braking systems using coordinate measuring machines; analyzed and reported deviations from nominal specifications.
- Tested quality control instruments and ensured they met tolerance parameters through detailed reporting.

Education

Master of Science in Computer Science (2026 – Present)

Georgia Institute of Technology. OMSCS Program. Specialization in Artificial Intelligence.

Mechatronics Engineering Licentiate Degree (2011 – 2016)

Costa Rica Institute of Technology. Graduated with honors. GPA: 90.51/100

Electronics & Technology Management. Exchange program supported by the Ernst-Mach scholarship (2014)

Fachhochschule Joanneum, Austria

Languages

- Spanish: Native language.
- English: Advanced level.
- German: Intermediate level.

Technical Skills

Backend: C/C++, C#, VB.NET, LabVIEW, Python, JavaScript, Matlab, Node.js, SQL.

Frontend: HTML/CSS, React, Blazor.

Databases: MySQL Server, Qdrant.

Software tools: Git, SourceTree, IBM ClearCase, Azure DevOps, Visual Studio, VS Code, Matlab, MySQL, LabVIEW, LabWindows/CVI, Measurement Studio, TestStand, FlexLogger, SystemLink, DIAdem, Docker, xUnit.